

Lucas Swierad

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Education

University of Michigan

BSE in Computer Science and Robotics

Expected Dec 2028

Ann Arbor, Michigan

- **Relevant Coursework:** Data Structures and Algorithms (C++), Intro to Machine Learning (Python), Web Systems, Intro to ROS, Deep Learning For Robotics, Intro to Computer Vision, Kinematics and Dynamics

Experience

Barracuda Networks

Site Reliability Engineering Intern

June 2026 – Present

Ann Arbor, Michigan

- Successfully decommissioned 100% of legacy Jenkins infrastructure across Barracuda Networks by migrating 14 active development pipelines to GitHub Actions.
- Secured sensitive pipeline data by integrating managed identities and Azure Key Vault into deployment workflows, automating the runtime retrieval of secrets and cryptographic keys to guarantee zero-exposure storage.

UM-CAEN

Multidisciplinary Design Program Intern

Jan 2026 – Dec 2026

Ann Arbor, Michigan

- Designing and developing a modular web application to automate administrative workflows for faculty and staff, reducing time spent on calendar management, file organization, and routine notifications.
- Implemented secure backend services using Node.js to support workflow automation, business logic, and API orchestration.
- Integrated Google Workspace APIs (Calendar, Drive, People) and Google OAuth / U-M SSO to enable permission-aware access, calendar automation, and metadata-driven file management.

UM Robotics

Teachers Assistant

Jan 2026 – May 2026

Ann Arbor, Michigan

- Assist with course instruction, including preparation of instructional materials and leading laboratory sections.
- Support lectures, recitation sections, and assessment through grading and exam administration.

Technical Skills

Languages: Python, C++, C, JavaScript, SQL

Technologies: SolidWorks, Onshape, GIT, ROS, PyTorch, Azure, AWS, Numpy, Scikit-Learn, OpenCV,

Projects

Self-Balancing Omnidirectional Ball-Bot | Onshape, Python, PID Control, 3D Printing, Kinematics

- Designed and implemented a PID-based closed-loop balance and steering control architecture for a 3-omni-wheel holonomic robot operating on a spherical surface.
- Built a multi-threaded Python control pipeline executing at 100 Hz over LCM (Lightweight Communications and Marshalling) to handle inter-process communication between a Raspberry Pi and Raspberry Pi Pico.

Dynamic Scene Reconstruction via Soft Mask Gating in 4DGS | PyTorch, 4DGS, GroundingDINO, SAM 2, NN

- Extended the official 4DGS codebase by appending a learnable Sigmoid-gated scalar branch to modulate multi-resolution HexPlane deformation feature vectors.
- Conducted comparative benchmark evaluations across D-NeRF and HyperNeRF datasets, analyzing trade-offs between rendering fidelity and spatial motion-mask separability.

Extracurricular

Autonomous Robotic Vehicle | PyTorch, Numpy, Pandas, LLMs, ChatGPT

- Developed and evaluated computer vision pipelines for autonomous navigation, including object detection, lane/feature recognition, and environment perception.
- Applied machine learning models to interpret sensor data (camera and IMU), improving robustness of real-time perception under varying environmental conditions.